SSL/X.509

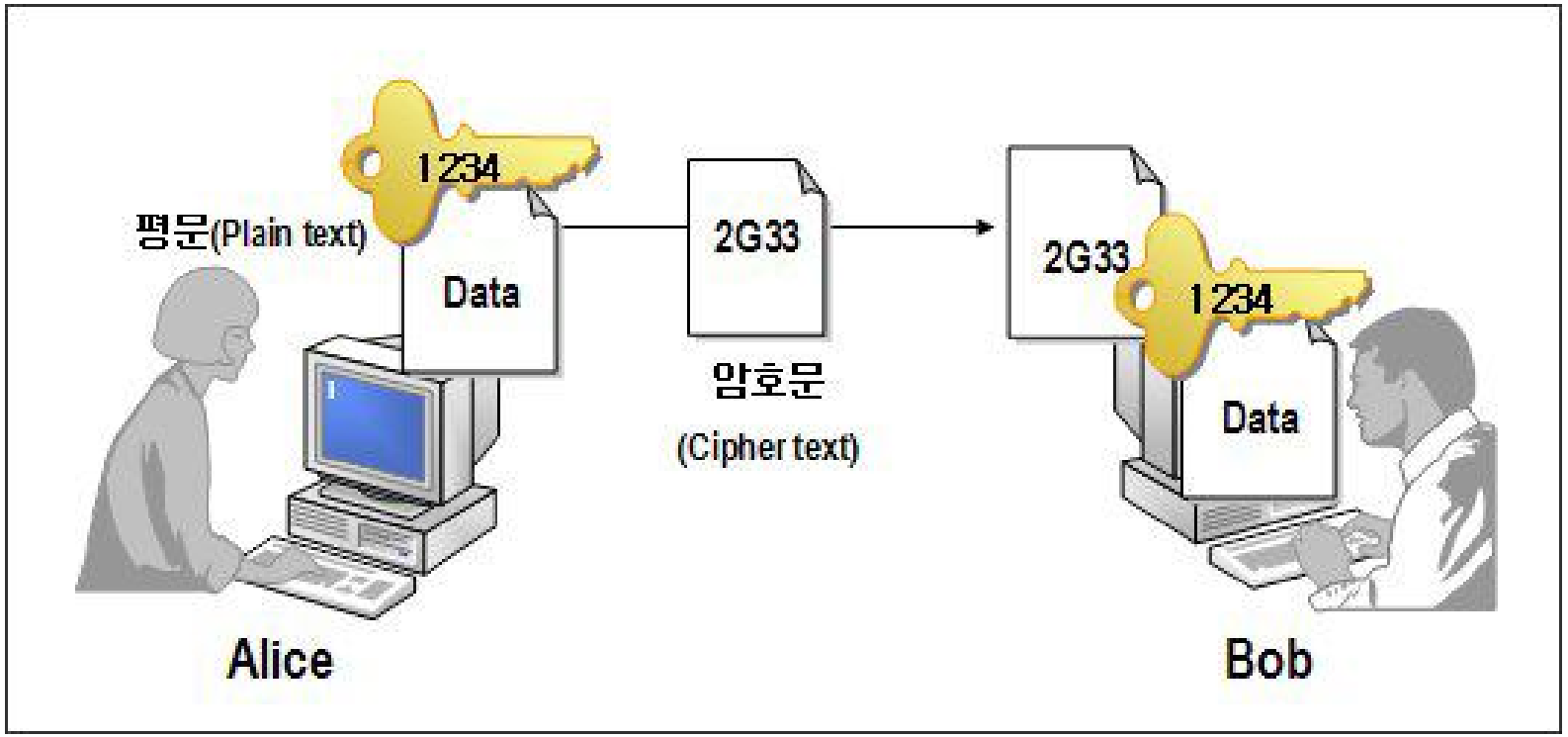
암호화의 역할

- Confidentiality (기밀성)
- Authenticity (신뢰성)
- Integrity (무결성)

Key를 이용한 암호화 기술로서 해결

❶ Symmetric Key(대칭키)

- Encryption Key = Decryption Key

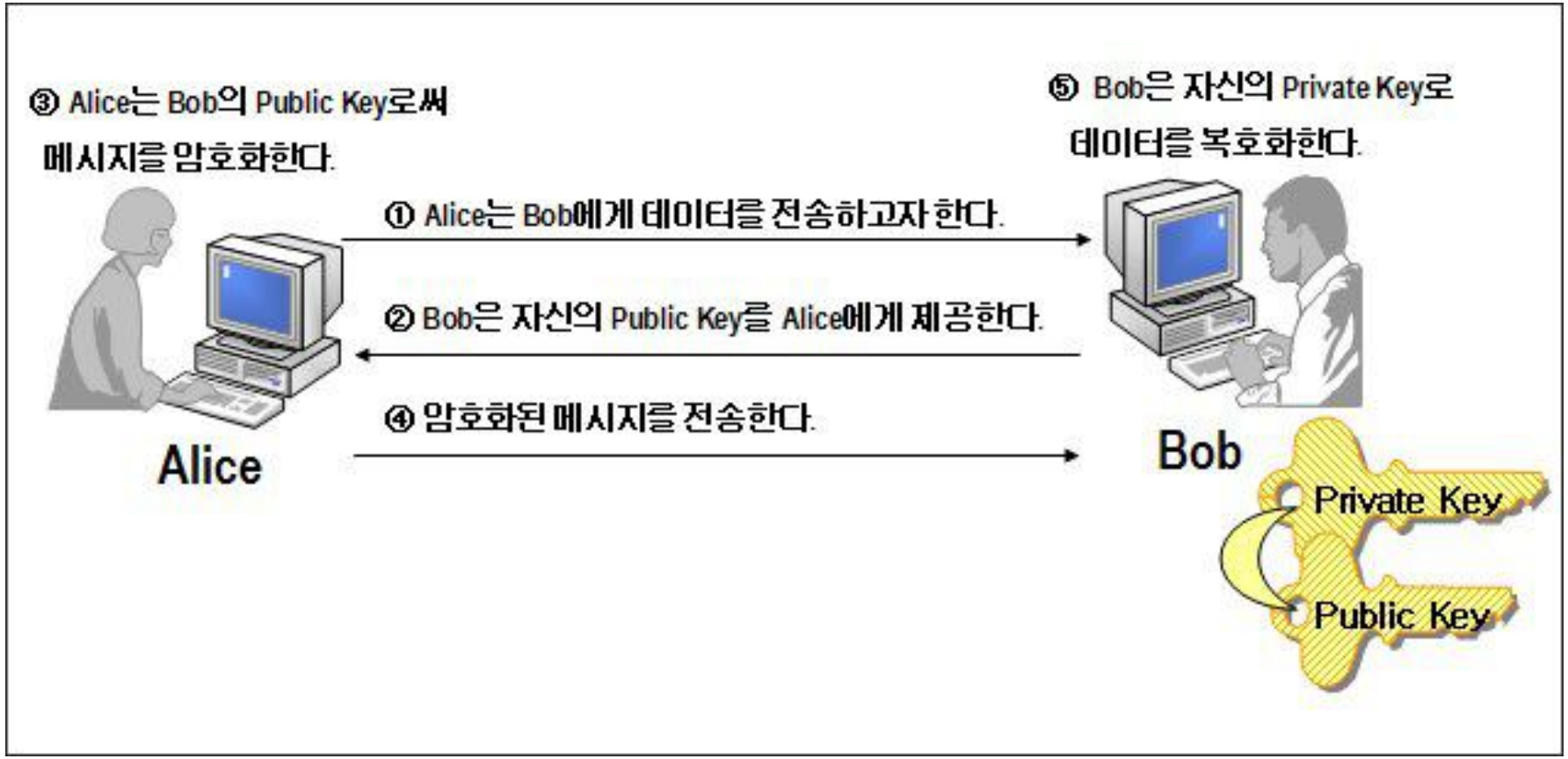


- DES, 3DES, AES, SEED, ARIA, RC5 알고리즘 등이 해당됨

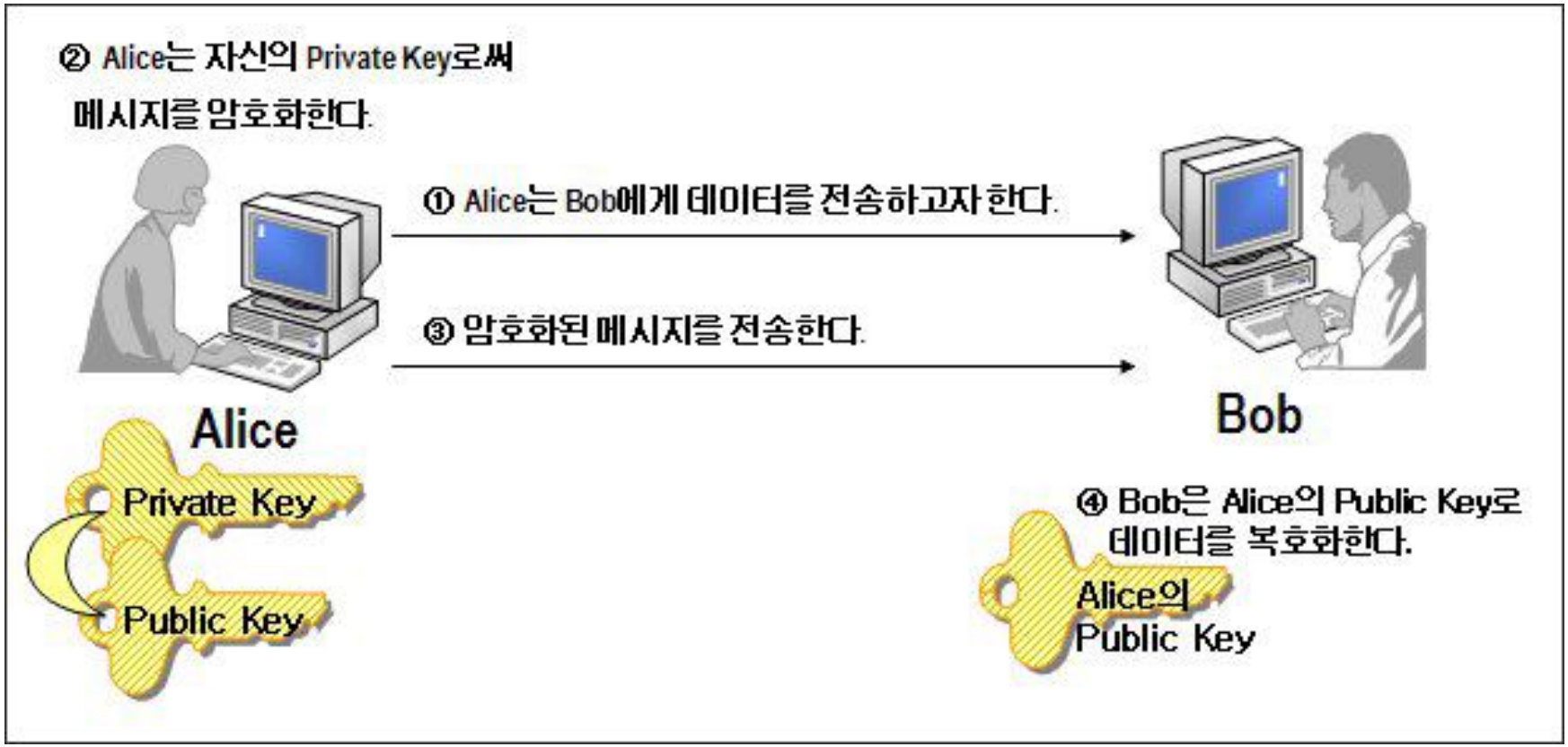
❷ Public Key(공용키)

- Encryption Key ≠ Decryption Key
  - 비대칭키, 공용키, 페어(Pair)키에 해당함.
- Public Key와 Private Key로 구성
  - Public Key(공용키) = 공개가 되는 키
  - Private Key(개인키) = 오직 발행주체만이 가지는 비밀키
- RSA 프로토콜

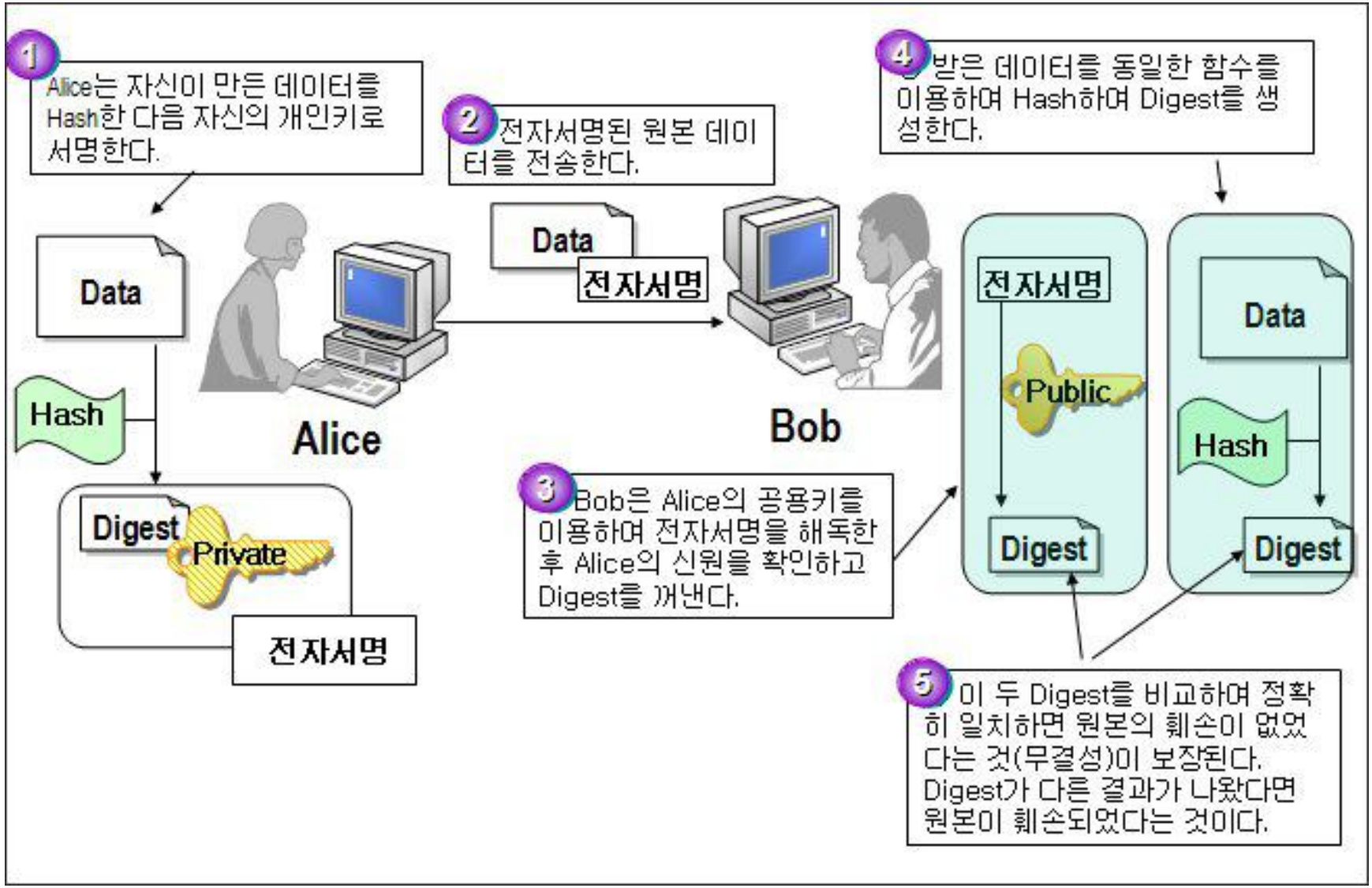
❸ Public Key암호화(Confidentiality제공)



❹ Private Key암호화(Authenticity제공)



❺ Hash Function & Digest (Integrity제공)

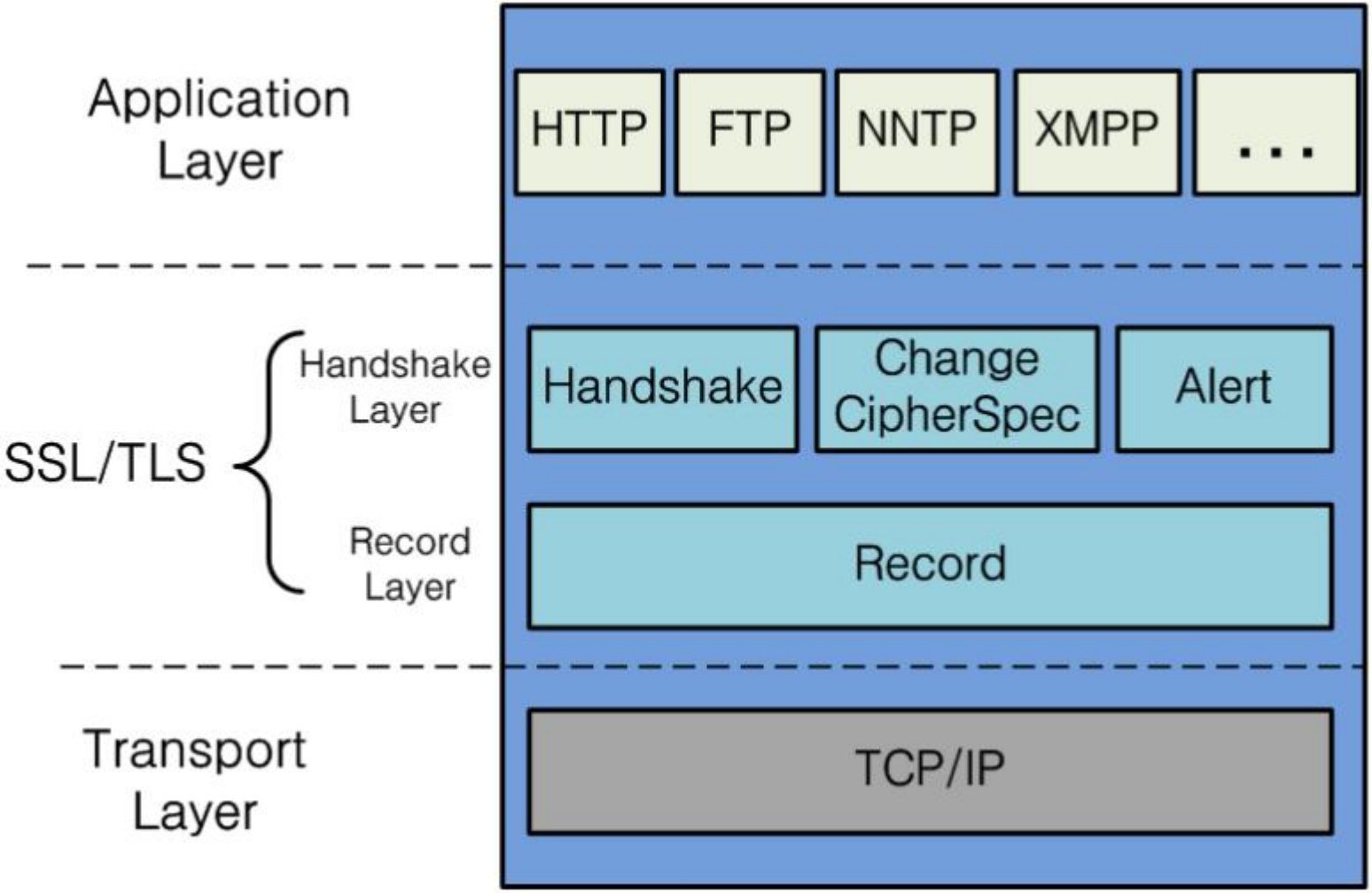


SSL/TLS

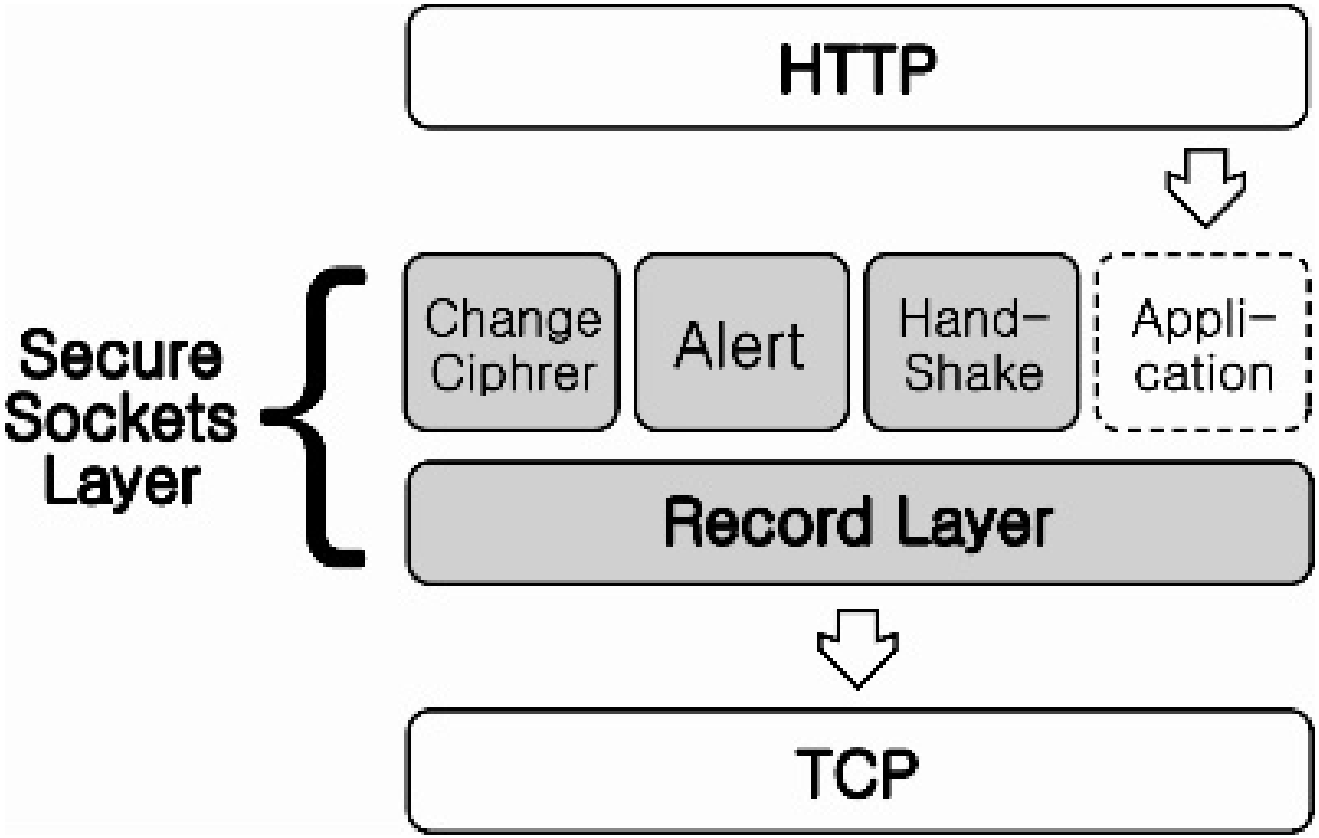
- 응용계층을 보호하는 프로토콜
- 웹 브라우저에 기본적으로 탑재되어 있음
- IETF에서 TLS 1.0 발표, SSL 3.1 이라고도 함
- TLS 프로토콜은 SSL 프로토콜을 사용할 수 있도록 구성
- SSL/TLS 프로토콜이라는 용어 사용

SSL(Secure Socket Layer)	TLS(Transport Layer Security)
응용계층을 보호하는 프로토콜	전송 계층 상위에서 동작하여 응용계층 암호화 HTTP 패킷 보호가 주용도

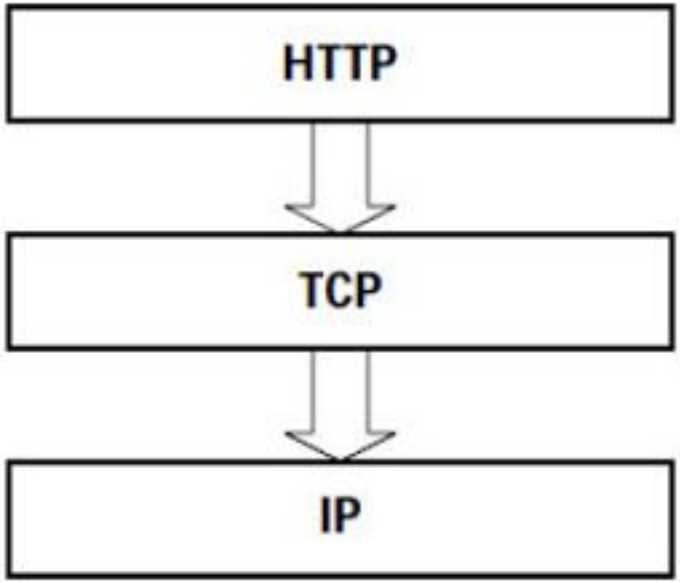
SSL/TLS Protocol stack



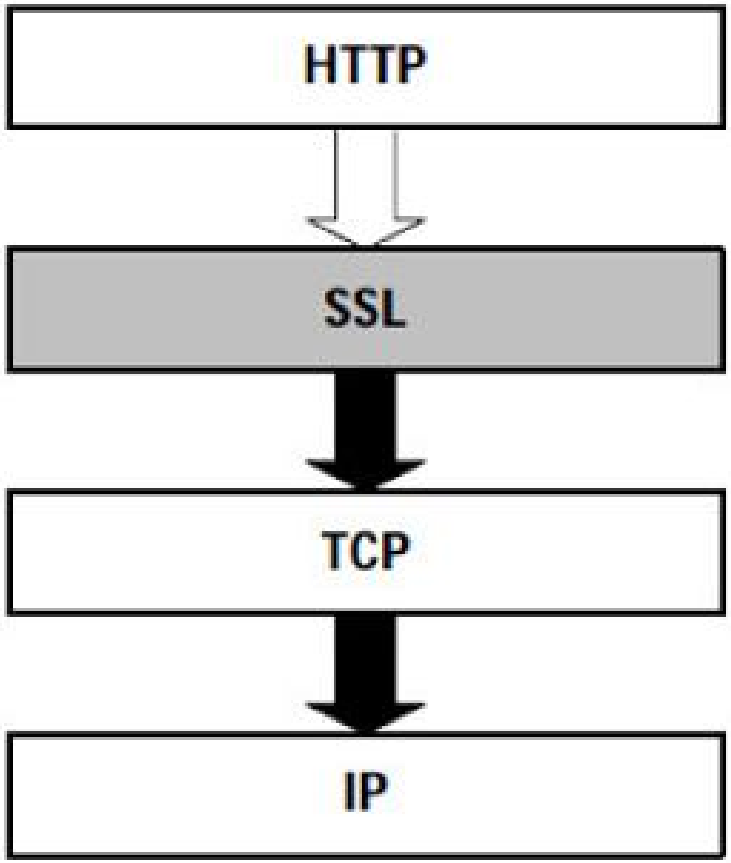
SSL/TLS Protocol stack



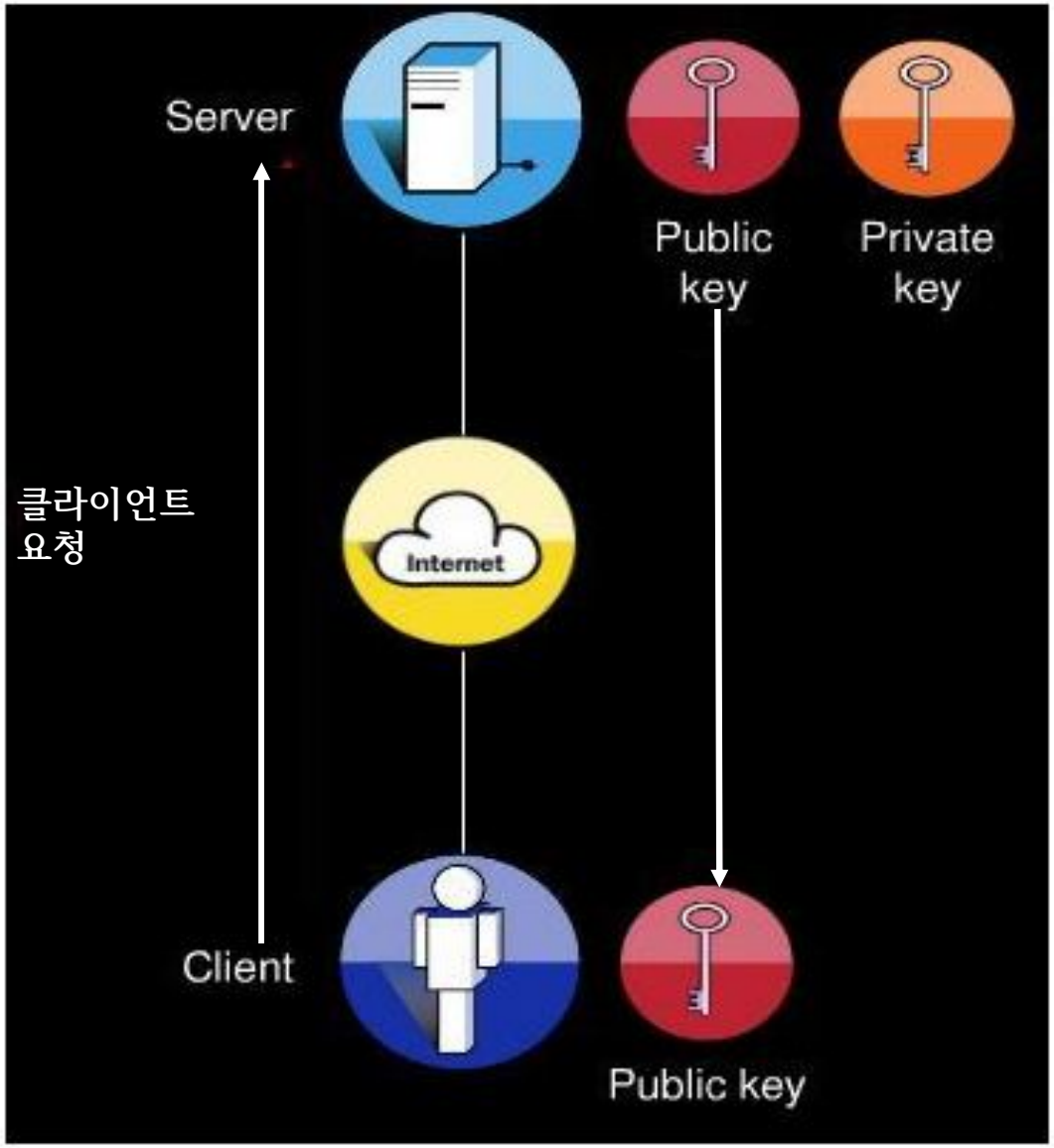
HTTP



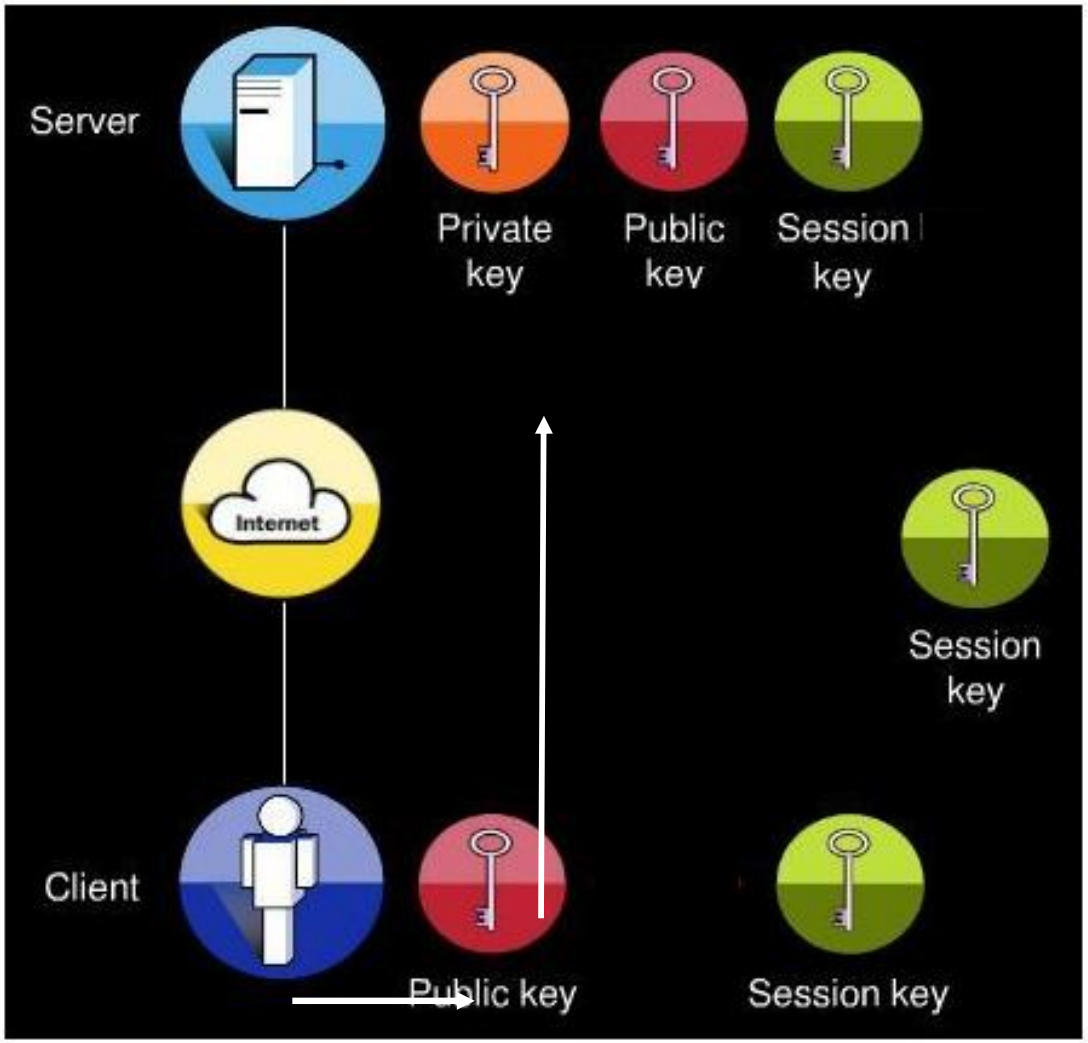
HTTPS



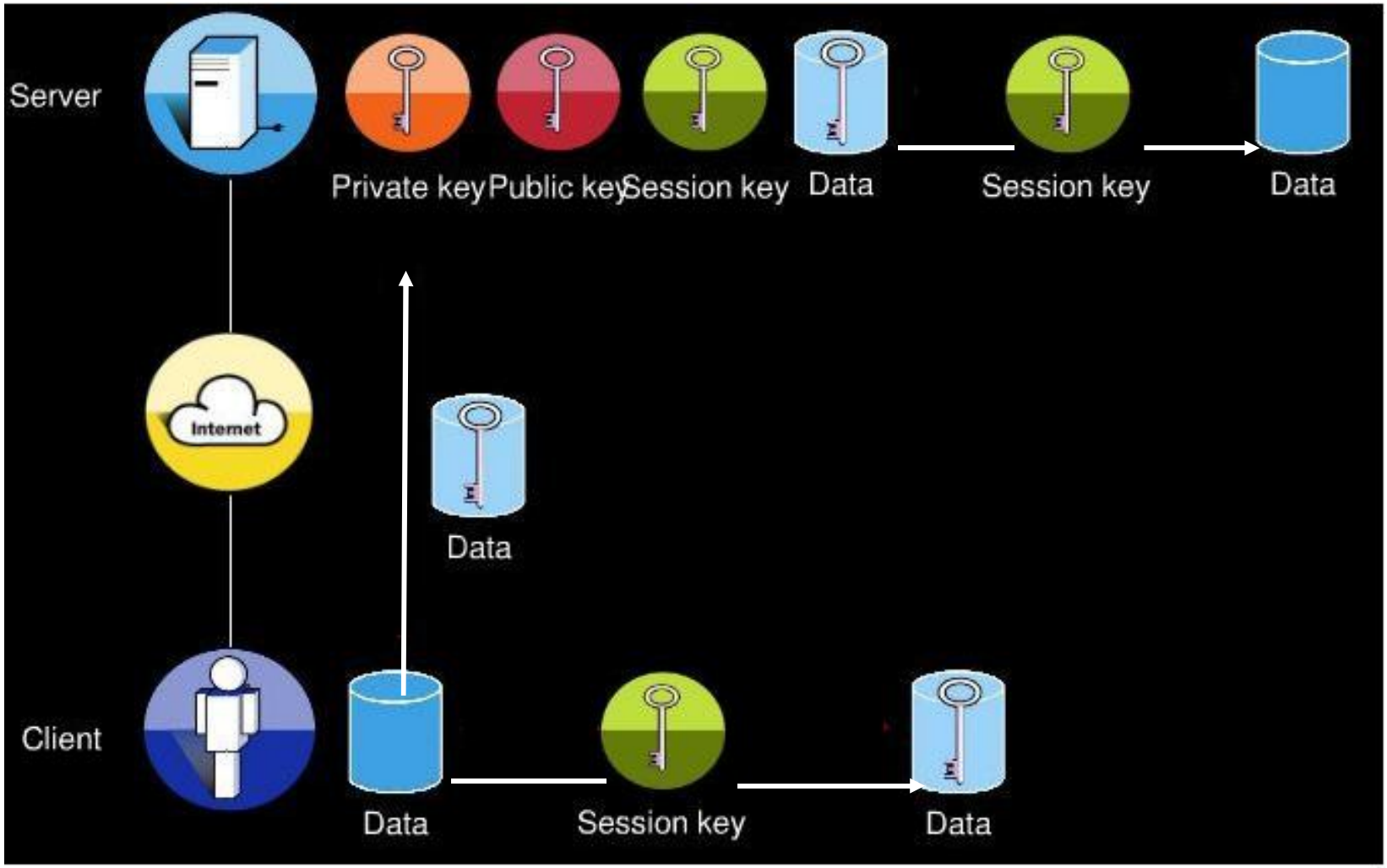
SSL/TLS Processing



Client가 server에 접속하면 서버인증서를 전송 받는다.

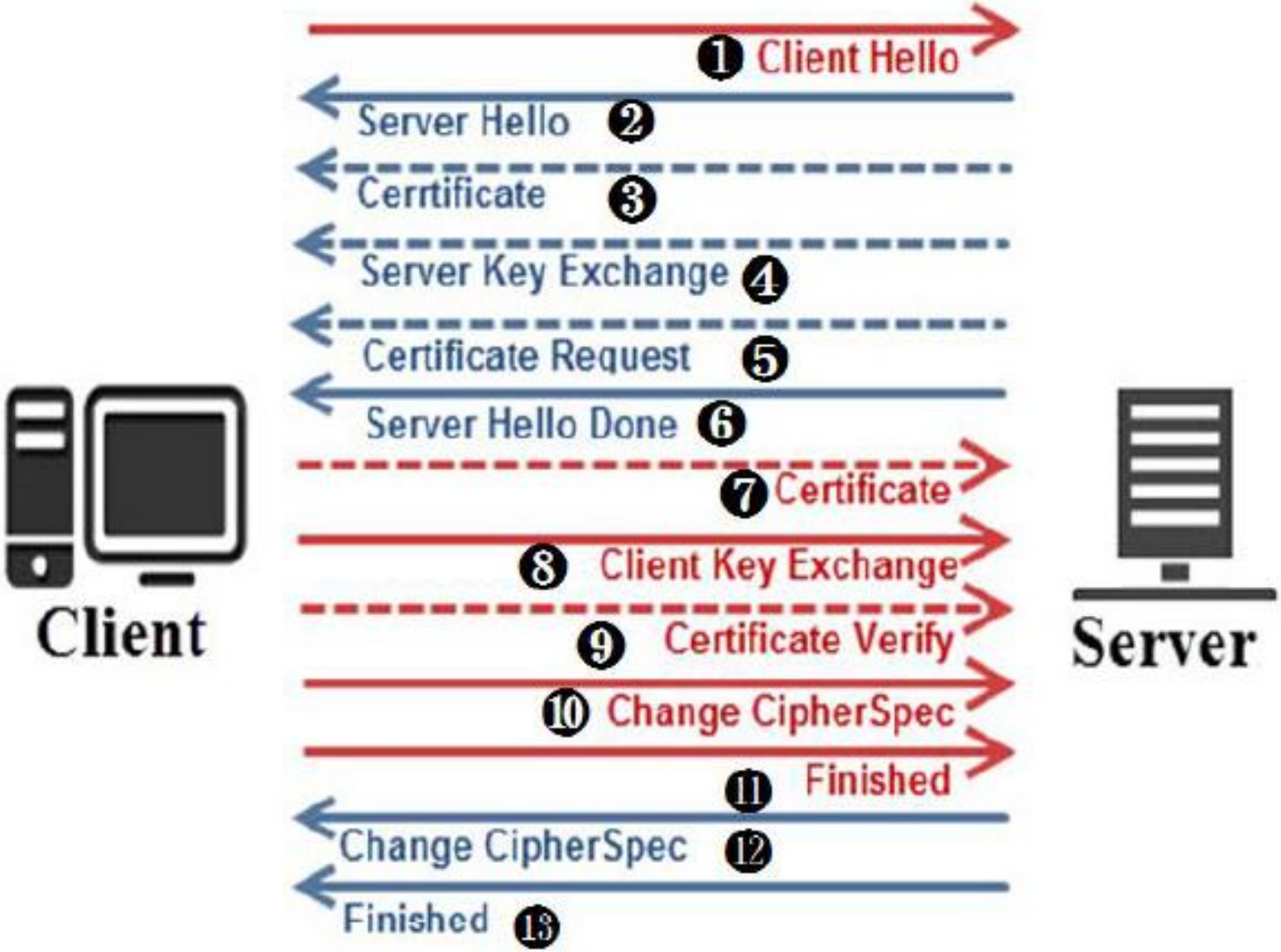


- Client는 받은 서버 인증서를 분석하여 신뢰할 수 있는 인증서 인지 검토
- 서버의 공개키를 추출
- Client가 세션키로 사용할 임의의 메시지는 서버의 공개키로 암호화하여 서버에 전송



- 서버에서는 자신의 개인키로 세션키를 복호화
- 세션키를 사용하여 대칭키 암호 방식으로 메세지를 암호화하여 클라이언트와 통신

  - Pre-Master Secret Key : 암호화된 세션키
  - Master Secret key : 복호화된 세션키


Client
Server
1 Client Hello
Server Hello 2
Cerrtificate 3
Server Key Exchange 4
Certificate Request 5
Server Hello Done 6
7 Certificate
8 Client Key Exchange
9 Certificate Verify
10 Change CipherSpec
11 Finished
Change CipherSpec 12
Finished 13

facebook_login.pcap

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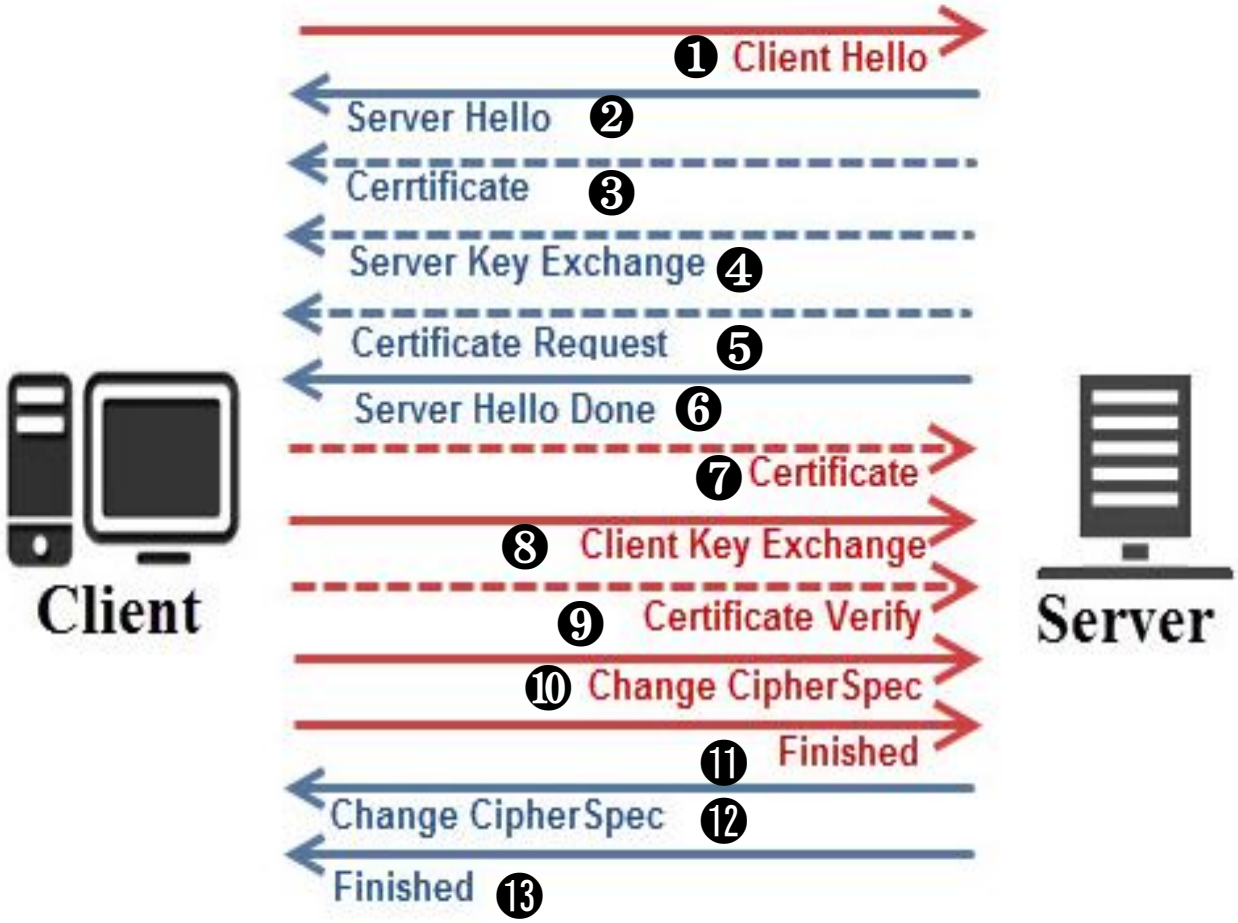
Apply a display filter ··· <Ctrl-/> Expression···

No.	Time	Source	Destination	Protocol	Total Length	Info
1	0.000000s	172.16.0.122	69.63.180.173	TCP	60	54595 → 443 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3019897…
2	0.089900s	69.63.180.173	172.16.0.122	TCP	64	443 → 54595 [SYN, ACK] Seq=0 Ack=1 Win=4140 Len=0 MSS=1380 WS=1 TSval=347…
3	0.000033s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=301989735 TSecr=347912…
4	0.000343s	172.16.0.122	69.63.180.173	TLSv1	221	Client Hello
5	0.089522s	69.63.180.173	172.16.0.122	TLSv1	989	Server Hello, Certificate, Server Hello Done
6	0.000031s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=170 Ack=938 Win=7744 Len=0 TSval=301989758 TSecr=34…
7	0.002848s	172.16.0.122	69.63.180.173	TLSv1	234	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
8	0.090444s	69.63.180.173	172.16.0.122	TLSv1	95	Change Cipher Spec, Encrypted Handshake Message
9	0.000533s	172.16.0.122	69.63.180.173	TLSv1	1034	Application Data
10	0.189619s	69.63.180.173	172.16.0.122	TCP	52	443 → 54595 [ACK] Seq=981 Ack=1334 Win=5473 Len=0 TSval=3479126142 TSecr=…
11	0.073201s	69.63.180.173	172.16.0.122	TLSv1	1233	Application Data
12	0.011497s	172.16.0.122	69.63.190.22	HTTP	679	GET /home.php? HTTP/1.1



ssl

No.	Time	Source	Destination	Protocol	Total Length	Info
4	0.000000s	172.16.0.122	69.63.180.173	TLSv1	221	Client Hello
5	0.089522s	69.63.180.173	172.16.0.122	TLSv1	989	Server Hello, Certificate, Server Hello Done
7	0.002879s	172.16.0.122	69.63.180.173	TLSv1	234	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
8	0.090444s	69.63.180.173	172.16.0.122	TLSv1	95	Change Cipher Spec, Encrypted Handshake Message
9	0.000533s	172.16.0.122	69.63.180.173	TLSv1	1034	Application Data
11	0.262820s	69.63.180.173	172.16.0.122	TLSv1	1233	Application Data
63	5m 1.502778s	172.16.0.122	69.63.180.173	TLSv1	75	Encrypted Alert

1 단계. Client Hello message

```
▷ Frame 4: 235 bytes on wire (1880 bits), 235 bytes captured (1880 bits)
▷ Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: Cisco_31:07:33 (00:26:0b:31:07:33)
▷ Internet Protocol Version 4, Src: 172.16.0.122, Dst: 69.63.180.173
▷ Transmission Control Protocol, Src Port: 54595, Dst Port: 443, Seq: 1, Ack: 1, Len: 169
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Handshake Protocol: Client Hello
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 164
    ◢ Handshake Protocol: Client Hello
        Handshake Type: Client Hello (1)
        Length: 160
        Version: TLS 1.0 (0x0301)
      ▷ Random: 4bba350339dc8387b20a0c5cfa490f4807d25f05c6c4cbdc...
        Session ID Length: 0
        Cipher Suites Length: 70
      ▷ Cipher Suites (35 suites)
        Compression Methods Length: 1
      ▷ Compression Methods (1 method)
        Extensions Length: 49
      ▷ Extension: server_name (len=23)
      ▷ Extension: supported_groups (len=8)
      ▷ Extension: ec_point_formats (len=2)
      ▷ Extension: SessionTicket TLS (len=0)
```

content

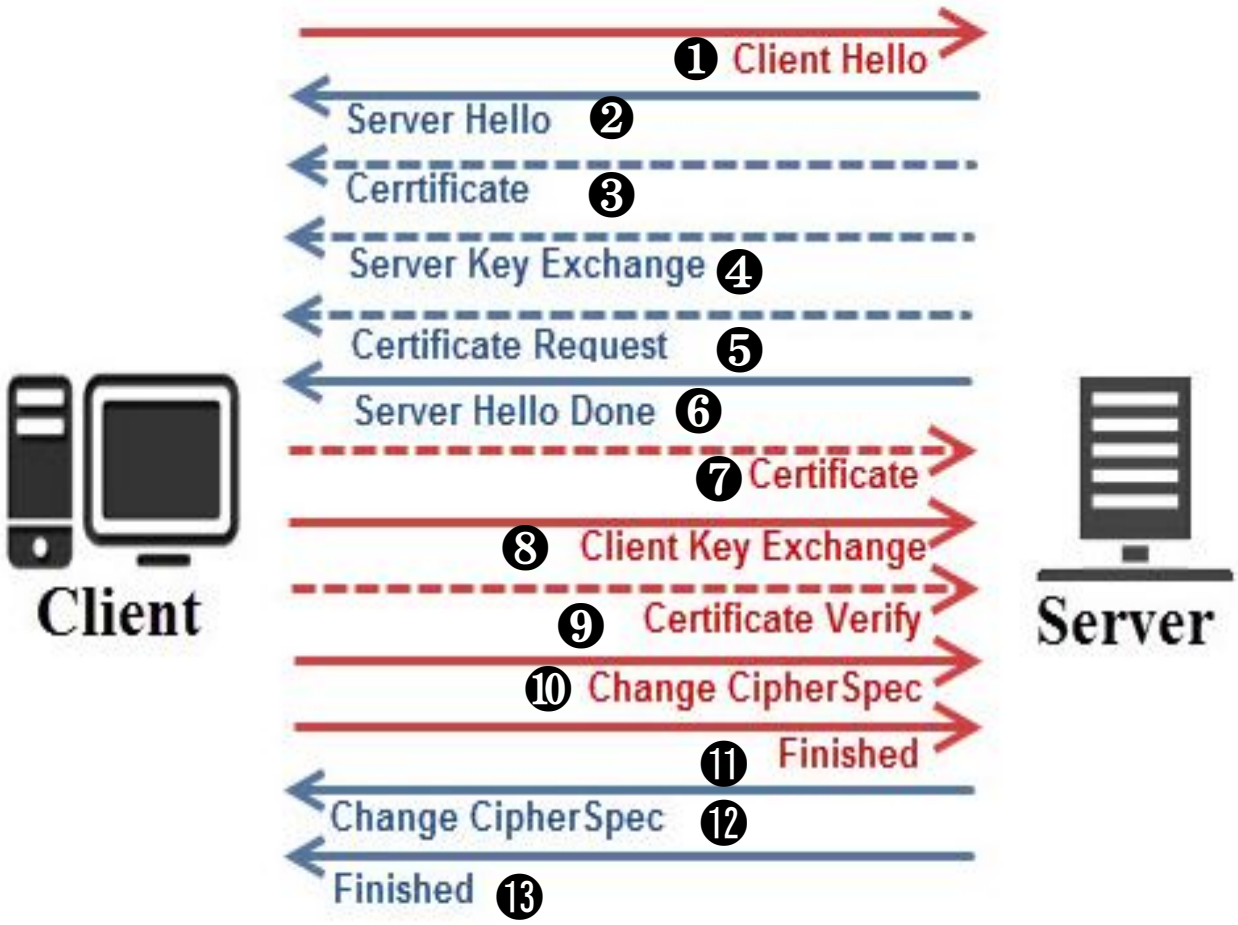
Protocol Suite

```
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Handshake Protocol: Client Hello
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 164
    ◢ Handshake Protocol: Client Hello
        Handshake Type: Client Hello (1)
        Length: 160
        Version: TLS 1.0 (0x0301)
      ◢ Random: 4bba350339dc8387b20a0c5cfa490f4807d25f05c6c4cbdc...
          GMT Unix Time: Apr  6, 2010 04:07:47.000000000 대한민국 표준시
          Random Bytes: 39dc8387b20a0c5cfa490f4807d25f05c6c4cbdc71fa59e8...
        Session ID Length: 0
        Cipher Suites Length: 70
      ◢ Cipher Suites (35 suites)
          Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA (0xc00a)
          Cipher Suite: TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (0xc014)
          Cipher Suite: TLS_DHE_RSA_WITH_CAMELLIA_256_CBC_SHA (0x0088)
          Cipher Suite: TLS_DHE_DSS_WITH_CAMELLIA_256_CBC_SHA (0x0087)
          Cipher Suite: TLS_DHE_RSA_WITH_AES_256_CBC_SHA (0x0039)
          Cipher Suite: TLS_DHE_DSS_WITH_AES_256_CBC_SHA (0x0038)
          Cipher Suite: TLS_ECDH_RSA_WITH_AES_256_CBC_SHA (0xc00f)
          Cipher Suite: TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA (0xc005)
          Cipher Suite: TLS_RSA_WITH_CAMELLIA_256_CBC_SHA (0x0084)
          Cipher Suite: TLS_RSA_WITH_AES_256_CBC_SHA (0x0035)
          Cipher Suite: TLS_ECDHE_ECDSA_WITH_RC4_128_SHA (0xc007)
```

```
    ▷ Random: 4bba350339dc8387b20a0c5cfa490f4807d25f05c6c4cbdc...
      Session ID Length: 0
      Cipher Suites Length: 70
    ▷ Cipher Suites (35 suites)
      Compression Methods Length: 1
    ◢ Compression Methods (1 method)
        Compression Method: null (0)
      Extensions Length: 49
    ◢ Extension: server_name (len=23)
        Type: server_name (0)
        Length: 23
      ▷ Server Name Indication extension
    ◢ Extension: supported_groups (len=8)
        Type: supported_groups (10)
        Length: 8
        Supported Groups List Length: 6
      ◢ Supported Groups (3 groups)
          Supported Group: secp256r1 (0x0017)
          Supported Group: secp384r1 (0x0018)
          Supported Group: secp521r1 (0x0019)
    ◢ Extension: ec_point_formats (len=2)
        Type: ec_point_formats (11)
        Length: 2
        EC point formats Length: 1
      ▷ Elliptic curves point formats (1)
    ◢ Extension: SessionTicket TLS (len=0)
        Type: SessionTicket TLS (35)
        Length: 0
        Data (0 bytes)
```

1 단계. Server Hello message

```
▷ Frame 5: 1003 bytes on wire (8024 bits), 1003 bytes captured (8024 bits)
▷ Ethernet II, Src: Cisco_31:07:33 (00:26:0b:31:07:33), Dst: Dell_c0:56:f0 (00:21:70:c0:56:f0)
▷ Internet Protocol Version 4, Src: 69.63.180.173, Dst: 172.16.0.122
▷ Transmission Control Protocol, Src Port: 443, Dst Port: 54595, Seq: 1, Ack: 170, Len: 937
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Handshake Protocol: Server Hello
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 74
    ◢ Handshake Protocol: Server Hello
        Handshake Type: Server Hello (2)
        Length: 70
        Version: TLS 1.0 (0x0301)
      ◢ Random: b9bb3b517aba70530291e8b0f97bb711647b94836658c94c...
          GMT Unix Time: Sep 28, 2068 19:56:17.000000000 대한민국 표준시
          Random Bytes: 7aba70530291e8b0f97bb711647b94836658c94c504630a2...
        Session ID Length: 32
        Session ID: 798e78f8199088e83fcf3e2ece32d14d26bc29eda5eb9149...
        Cipher Suite: TLS_RSA_WITH_RC4_128_MD5 (0x0004)
        Compression Method: null (0)
  ▷ TLSv1 Record Layer: Handshake Protocol: Certificate
  ▷ TLSv1 Record Layer: Handshake Protocol: Server Hello Done
```



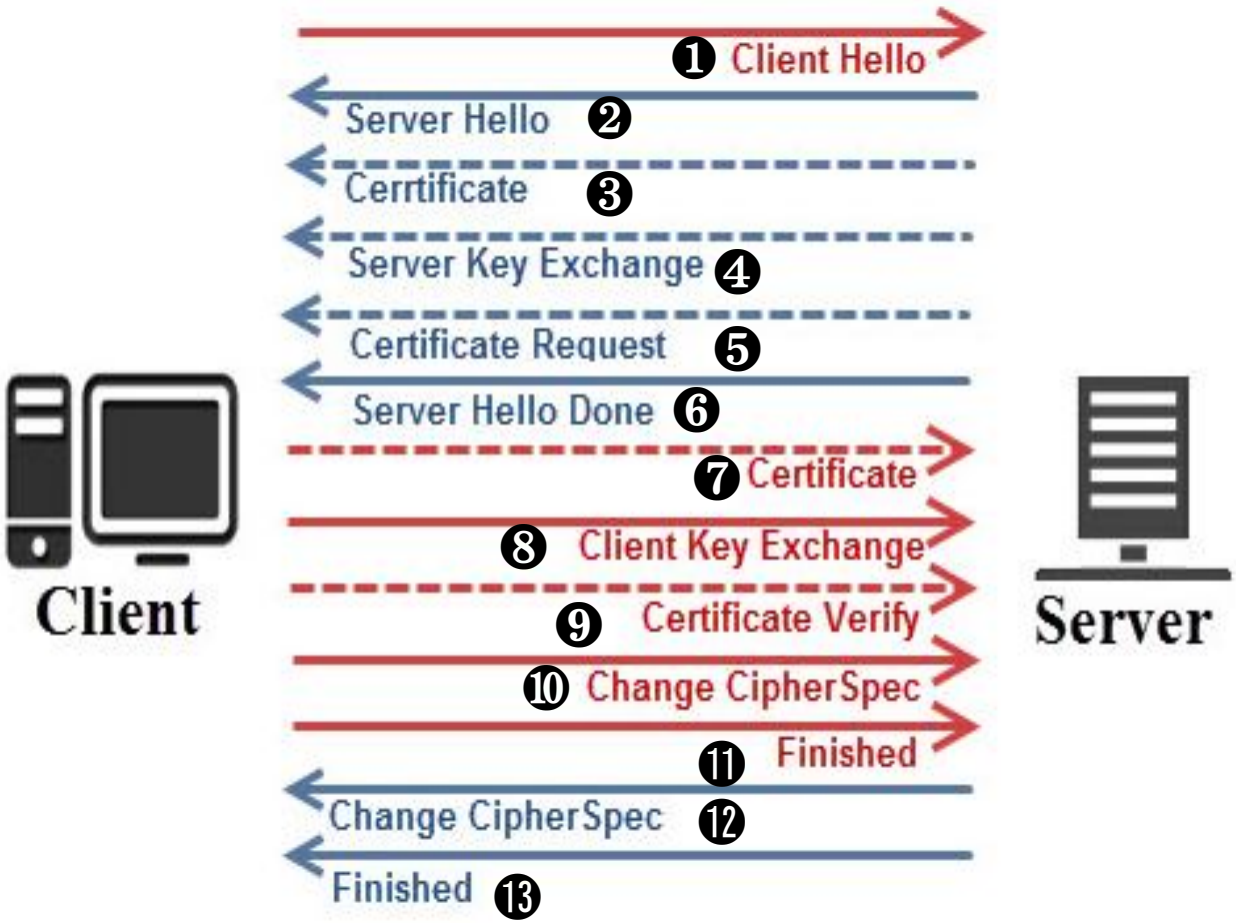

```
▷ Frame 5: 1003 bytes on wire (8024 bits), 1003 bytes captured (8024 bits)
▷ Ethernet II, Src: Cisco_31:07:33 (00:26:0b:31:07:33), Dst: Dell_c0:56:f0 (00:21:70:c0:56:f0)
▷ Internet Protocol Version 4, Src: 69.63.180.173, Dst: 172.16.0.122
▷ Transmission Control Protocol, Src Port: 443, Dst Port: 54595, Seq: 1, Ack: 170, Len: 937
◢ Secure Sockets Layer
  ▷ TLSv1 Record Layer: Handshake Protocol: Server Hello
  ▷ TLSv1 Record Layer: Handshake Protocol: Certificate
  ▷ TLSv1 Record Layer: Handshake Protocol: Server Hello Done
```

2 단계. Certificate message

```
▷ TLSv1 Record Layer: Handshake Protocol: Server Hello
◢ TLSv1 Record Layer: Handshake Protocol: Certificate
    Content Type: Handshake (22)
    Version: TLS 1.0 (0x0301)
    Length: 844
  ◢ Handshake Protocol: Certificate
      Handshake Type: Certificate (11)
      Length: 840
      Certificates Length: 837
    ◢ Certificates (837 bytes)
        Certificate Length: 834
      ◢ Certificate: 3082033e308202a7a00302010202030c183f300d06092a86... (id-at-commonName=login.facebook.com,id-at-organizationalU
        ◢ signedCertificate
            version: v3 (2)
            serialNumber: 792639
          ▷ signature (sha1WithRSAEncryption)
          ▷ issuer: rdnSequence (0)
          ▷ validity
          ▷ subject: rdnSequence (0)
          ◢ subjectPublicKeyInfo
            ◢ algorithm (rsaEncryption)
                Algorithm Id: 1.2.840.113549.1.1.1 (rsaEncryption)
            ▷ subjectPublicKey: 30818902818100d6114e489ea6f3a3611ac38d0b3e46a30c...
          ▷ extensions: 5 items
        ◢ algorithmIdentifier (sha1WithRSAEncryption)
            Algorithm Id: 1.2.840.113549.1.1.5 (sha1WithRSAEncryption)
          Padding: 0
          encrypted: 36104b5d5b134d9fc9b0b9a47d1131e528992f9a6bf2f74f...
```

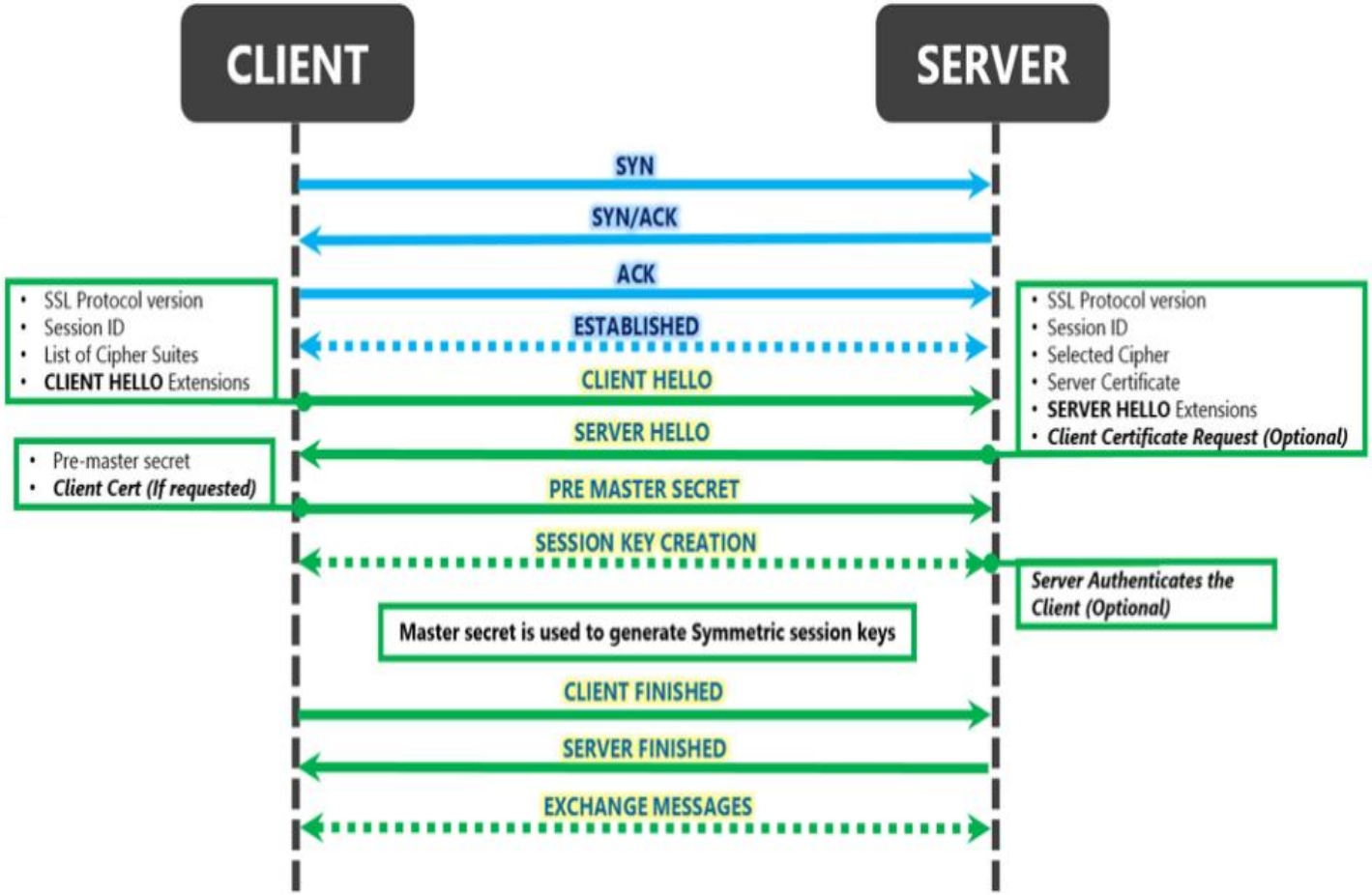
2 단계. Server Hello Done message

```
▷ Frame 5: 1003 bytes on wire (8024 bits), 1003 bytes captured (8024 bits)
▷ Ethernet II, Src: Cisco_31:07:33 (00:26:0b:31:07:33), Dst: Dell_c0:56:f0 (00:21:70:c0:56:f0)
▷ Internet Protocol Version 4, Src: 69.63.180.173, Dst: 172.16.0.122
▷ Transmission Control Protocol, Src Port: 443, Dst Port: 54595, Seq: 1, Ack: 170, Len: 937
◢ Secure Sockets Layer
  ▷ TLSv1 Record Layer: Handshake Protocol: Server Hello
  ▷ TLSv1 Record Layer: Handshake Protocol: Certificate
  ◢ TLSv1 Record Layer: Handshake Protocol: Server Hello Done
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 4
    ◢ Handshake Protocol: Server Hello Done
        Handshake Type: Server Hello Done (14)
        Length: 0
```



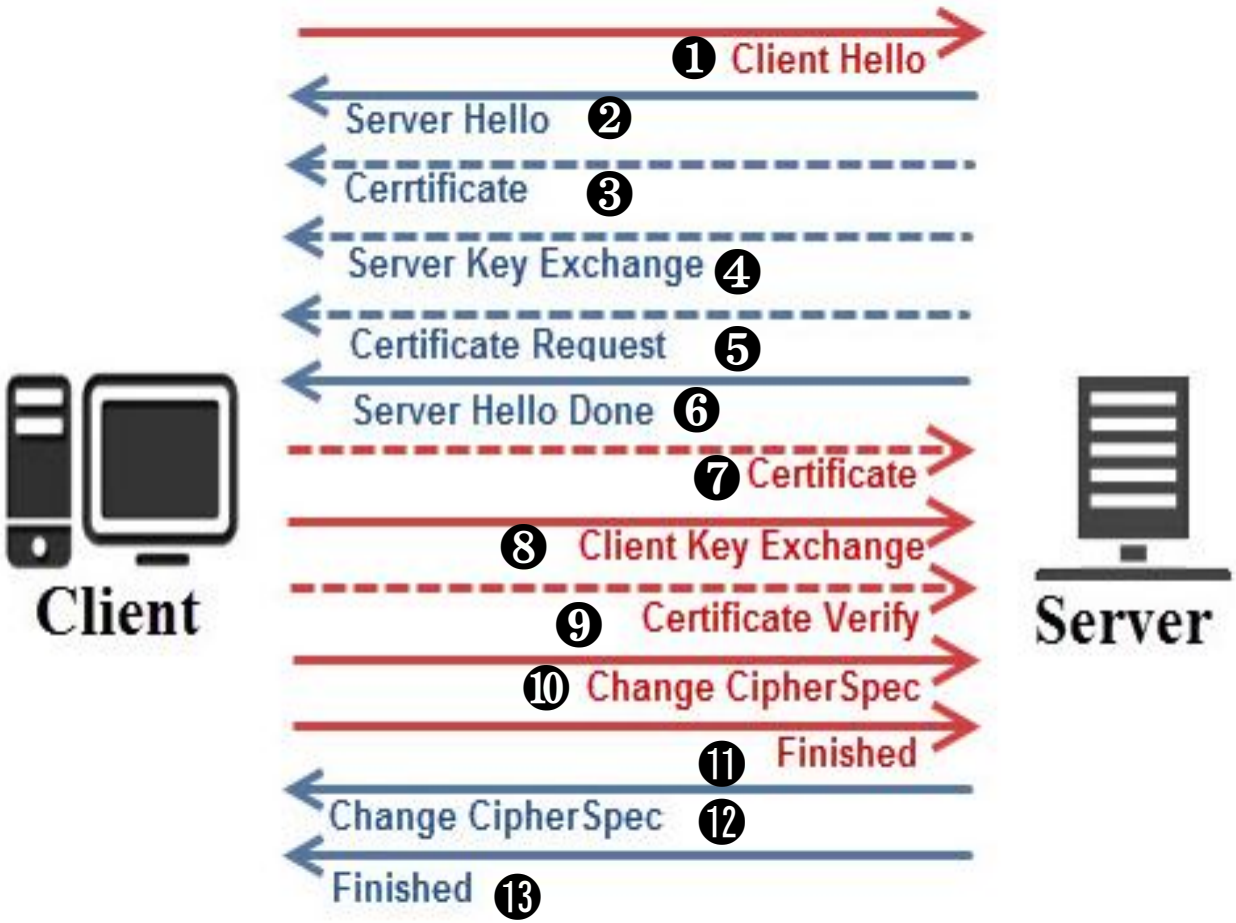
	Time	Source	Destination	Protocol	Total Length	Info
1	0.000000s	172.16.0.122	69.63.180.173	TCP	60	54595 → 443 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3019897…
2	0.089900s	69.63.180.173	172.16.0.122	TCP	64	443 → 54595 [SYN, ACK] Seq=0 Ack=1 Win=4140 Len=0 MSS=1380 WS=1 TSval=347…
3	0.000033s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=301989735 TSecr=347912…
4	0.000343s	172.16.0.122	69.63.180.173	TLSv1	221	Client Hello
5	0.089522s	69.63.180.173	172.16.0.122	TLSv1	989	Server Hello, Certificate, Server Hello Done
6	0.000031s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=170 Ack=938 Win=7744 Len=0 TSval=301989758 TSecr=34…
7	0.002848s	172.16.0.122	69.63.180.173	TLSv1	234	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
8	0.090444s	69.63.180.173	172.16.0.122	TLSv1	95	Change Cipher Spec, Encrypted Handshake Message
9	0.000533s	172.16.0.122	69.63.180.173	TLSv1	1034	Application Data
10	0.189619s	69.63.180.173	172.16.0.122	TCP	52	443 → 54595 [ACK] Seq=981 Ack=1334 Win=5473 Len=0 TSval=3479126142 TSecr=…
11	0.073201s	69.63.180.173	172.16.0.122	TLSv1	1233	Application Data
12	0.011497s	172.16.0.122	69.63.190.22	HTTP	679	GET /home.php? HTTP/1.1

3 단계.




```
▷ Frame 7: 248 bytes on wire (1984 bits), 248 bytes captured (1984 bits)
▷ Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: Cisco_31:07:33 (00:26:0b:31:07:33)
▷ Internet Protocol Version 4, Src: 172.16.0.122, Dst: 69.63.180.173
▷ Transmission Control Protocol, Src Port: 54595, Dst Port: 443, Seq: 170, Ack: 938, Len: 182
◢ Secure Sockets Layer
  ▷ TLSv1 Record Layer: Handshake Protocol: Client Key Exchange
  ▷ TLSv1 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec
  ▷ TLSv1 Record Layer: Handshake Protocol: Encrypted Handshake Message
```

```
▷ Frame 7: 248 bytes on wire (1984 bits), 248 bytes captured (1984 bits)
▷ Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: Cisco_31:07:33 (00:26:0b:31:07:33)
▷ Internet Protocol Version 4, Src: 172.16.0.122, Dst: 69.63.180.173
▷ Transmission Control Protocol, Src Port: 54595, Dst Port: 443, Seq: 170, Ack: 938, Len: 182
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Handshake Protocol: Client Key Exchange
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 134
    ◢ Handshake Protocol: Client Key Exchange
        Handshake Type: Client Key Exchange (16)
        Length: 130
      ◢ RSA Encrypted PreMaster Secret
          Encrypted PreMaster length: 128
          Encrypted PreMaster: 3b68c9a6fea0f7888813d309c8a1d81344b4b01f17d9a8ec...
  ◢ TLSv1 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec
      Content Type: Change Cipher Spec (20)
      Version: TLS 1.0 (0x0301)
      Length: 1
      Change Cipher Spec Message
  ◢ TLSv1 Record Layer: Handshake Protocol: Encrypted Handshake Message
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 32
      Handshake Protocol: Encrypted Handshake Message
```



No.	Time	Source	Destination	Protocol	Total Length	Info
1	0.000000s	172.16.0.122	69.63.180.173	TCP	60	54595 → 443 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3019897…
2	0.089900s	69.63.180.173	172.16.0.122	TCP	64	443 → 54595 [SYN, ACK] Seq=0 Ack=1 Win=4140 Len=0 MSS=1380 WS=1 TSval=347…
3	0.000033s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=301989735 TSecr=347912…
4	0.000343s	172.16.0.122	69.63.180.173	TLSv1	221	Client Hello
5	0.089522s	69.63.180.173	172.16.0.122	TLSv1	989	Server Hello, Certificate, Server Hello Done
6	0.000031s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=170 Ack=938 Win=7744 Len=0 TSval=301989758 TSecr=34…
7	0.002848s	172.16.0.122	69.63.180.173	TLSv1	234	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
8	0.090444s	69.63.180.173	172.16.0.122	TLSv1	95	Change Cipher Spec, Encrypted Handshake Message
9	0.000533s	172.16.0.122	69.63.180.173	TLSv1	1034	Application Data
10	0.189619s	69.63.180.173	172.16.0.122	TCP	52	443 → 54595 [ACK] Seq=981 Ack=1334 Win=5473 Len=0 TSval=3479126142 TSecr=…
11	0.073201s	69.63.180.173	172.16.0.122	TLSv1	1233	Application Data
12	0.011497s	172.16.0.122	69.63.190.22	HTTP	679	GET /home.php? HTTP/1.1

```
▷ Frame 8: 109 bytes on wire (872 bits), 109 bytes captured (872 bits)
▷ Ethernet II, Src: Cisco_31:07:33 (00:26:0b:31:07:33), Dst: Dell_c0:56:f0 (00:21:70:c0:56:f0)
▷ Internet Protocol Version 4, Src: 69.63.180.173, Dst: 172.16.0.122
▷ Transmission Control Protocol, Src Port: 443, Dst Port: 54595, Seq: 938, Ack: 352, Len: 43
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec
      Content Type: Change Cipher Spec (20)
      Version: TLS 1.0 (0x0301)
      Length: 1
      Change Cipher Spec Message
  ◢ TLSv1 Record Layer: Handshake Protocol: Encrypted Handshake Message
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 32
      Handshake Protocol: Encrypted Handshake Message
```